

AD HOC NETWORKS

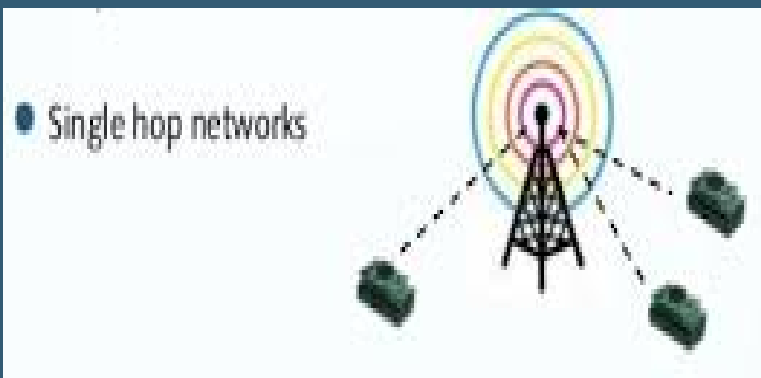
MANETS & WIRELESS SENSOR NETWORK

Course instructor:

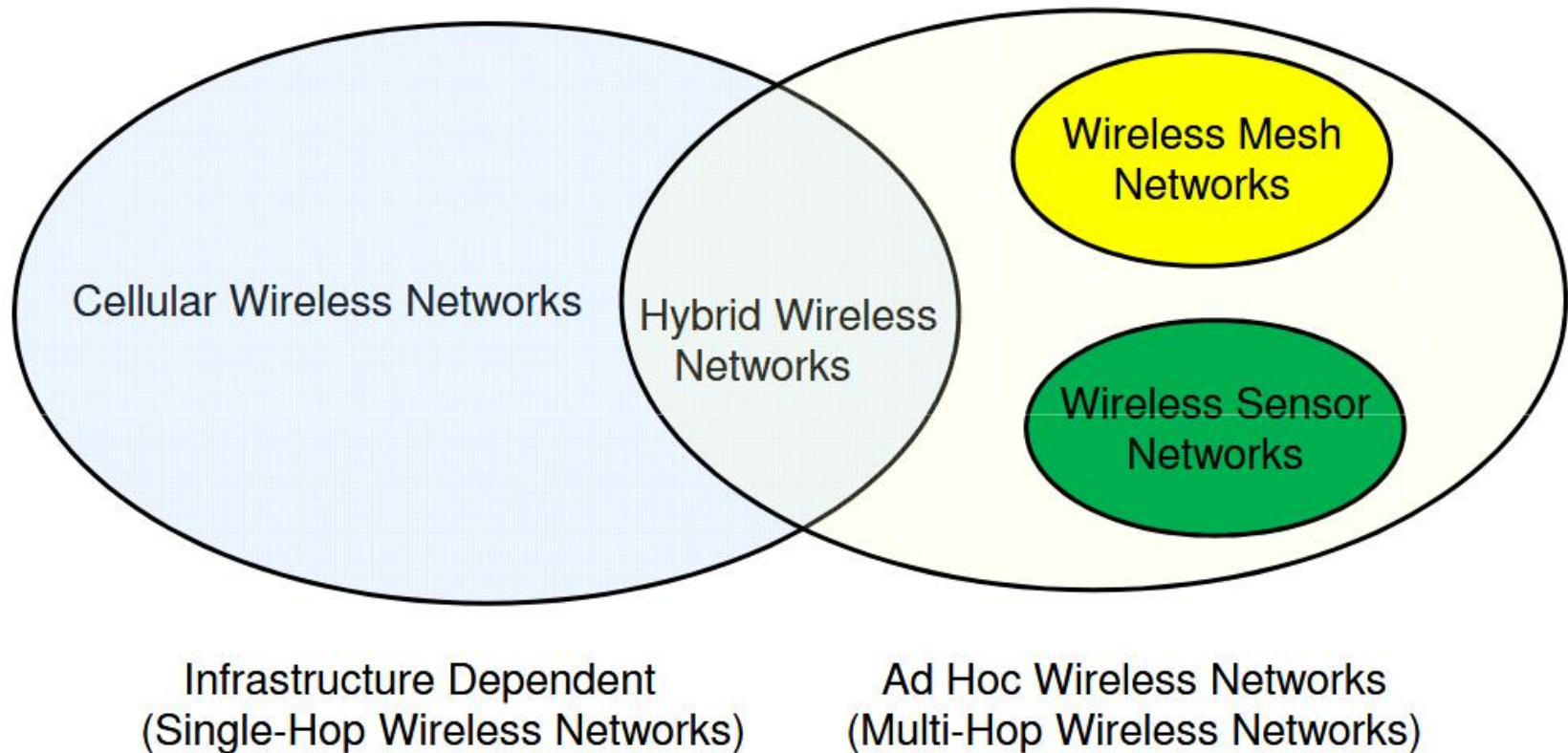
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AD HOC NETWORKS

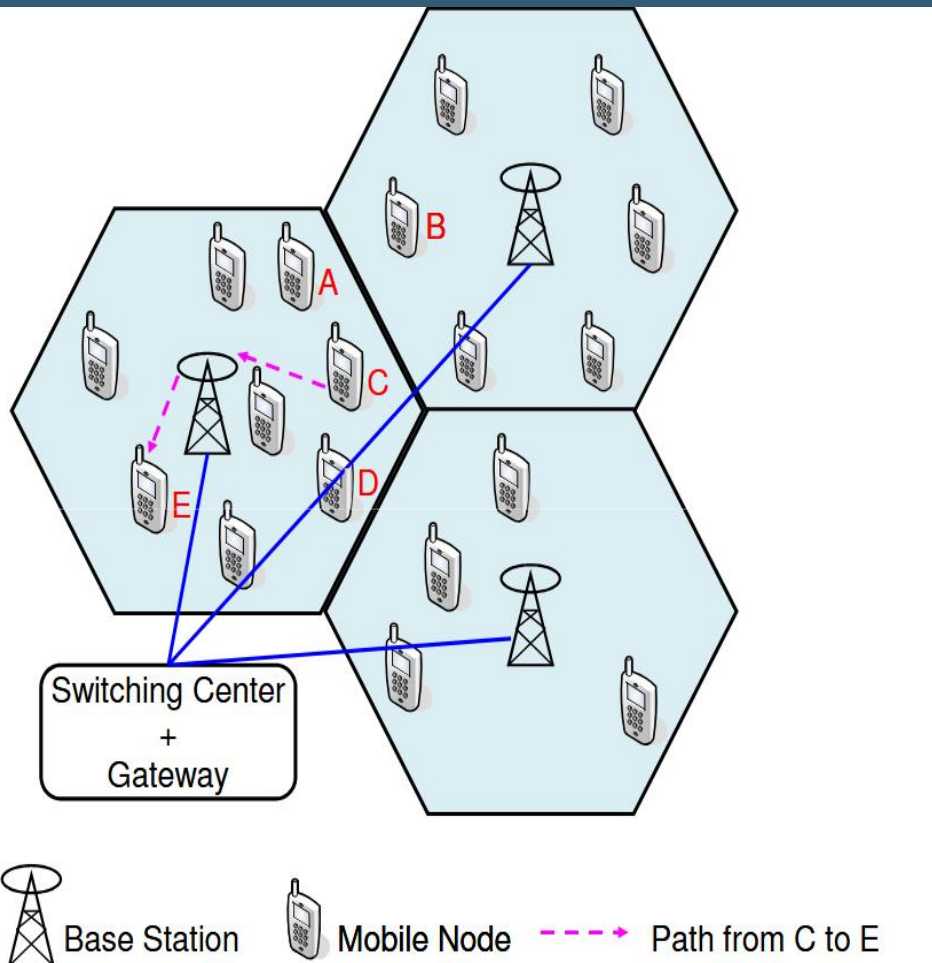
- An ad hoc network is a network that is composed of individual devices communicating with each other directly.
- Ad Hoc Network is a multi-hop relaying network.
- Multi-hop, or ad hoc, wireless networks use two or more wireless hops to convey information from a source to a destination



CLASSIFICATION OF WIRELESS NETWORKS

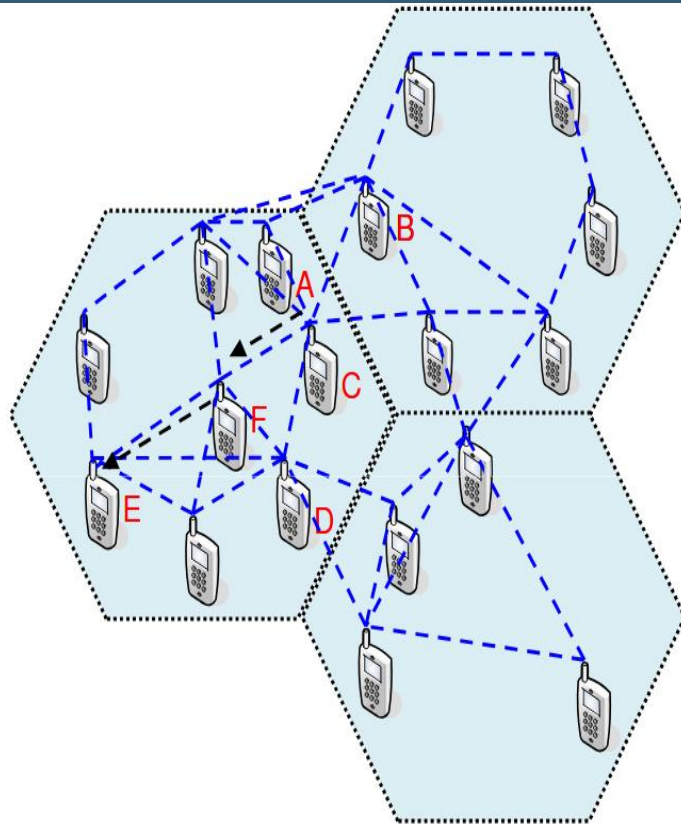


CELLULAR NETWORKS



- Infrastructure Dependent network
- Needs Base station for communication between two devices
- If communication is required from C to E then C-> Base station ->E
- Presence of base station simplifies the routing

AD HOC NETWORKS



Mobile Node

----- Wireless Link

-----> Path from C to E

- Infrastructure less Network
- Absence of base station make the routing complex
- If communication is required from C to E then A-> F ->E
- All nodes can enable the communication
- Every node can act as transmitter, receiver and router

DIFFERENCE BETWEEN CELLULAR AND AD HOC NETWORKS

Cellular networks

- Fixed infrastructure-based
- Single-hop wireless links
- Guaranteed bandwidth (designed for voice traffic)
- Centralized routing
- Circuit-switched (evolving toward packet switching)
- Seamless connectivity (low call drops during handoffs)
- High cost and time of deployment

Ad hoc networks

- Infrastructure-less
- Multi-hop wireless links
- Shared radio channel (more suitable for best-effort data traffic)
- Distributed routing
- Packet-switched (evolving toward emulation of circuit switching)
- Frequency path break due to mobility
- Quick and cost-effective deployment

DIFFERENCE BETWEEN CELLULAR AND AD HOC NETWORKS

Cellular networks

- Reuse of frequency spectrum through geographical channel reuse
- Easier to achieve time synchronization
- Easier to employ bandwidth reservation
- Application domains include mainly civilian and commercial sector
- High cost of network maintenance (backup power source, staffing, etc.)

Ad hoc networks

- Dynamic frequency reuse based on carrier sense mechanism
- Time synchronization is difficult and consumes bandwidth
- Bandwidth reservation requires complex medium access control protocols
- Application domains include battlefields, emergency search and rescue operation, and collaborative computing
- Self-organization and maintenance properties are built into the network

DIFFERENCE BETWEEN CELLULAR AND AD HOC NETWORKS

Cellular networks

- Mobile hosts are of relatively low complexity
- Major goals of routing and call admission are to maximize the call acceptance ratio and minimize the call drop ratio
- Widely deployed and currently in the third generation

Ad hoc networks

- Mobile hosts require more intelligence (should have a transceiver as well as routing/switching capacity)
- Main aim of routing is to find paths with minimum overhead and also quick reconfiguration of broken paths
- Several issues are to be addressed for successful commercial deployment even though widespread use exists in defense

APPLICATIONS OF AD HOC WIRELESS NETWORKS

- **Military Applications**

- Establishing communication among a group of soldiers for tactical operations
- Coordination of military object moving at high speeds such as fleets of airplanes or ships
- Requirements: reliability, efficiency, secure communication and multicast routing

- **Collaborative and Distributed Computing**

- Conference, distributed files sharing

- **Emergency Operations**

- Search, rescue, crowd control, and commando operations
- Support real-time and fault-tolerant communication paths

WIRELESS MESH NETWORK (WMN)

- A **Wireless Mesh Network (WMN)** is a type of ad hoc network where each node (device) can connect to **multiple other nodes**, creating a **mesh of wireless connections**.
- Every node acts as both a **host** and a **router**, forwarding data for the network.
- WMN is an alternate communication infrastructure for mobile or fixed nodes/users
- WMN provides many alternate paths for a data transfer session between a source and transfer session between a source and destination
- Advantages of Wireless Mesh Networks
 - High data rate, quick and low cost of deployment, enhanced services, high scalability, easy extendibility, high availability, and low cost per bit

TECHNOLOGIES USED FOR WMN

- 802.11s (Mesh Wi-Fi)
- Zigbee Mesh (for IoT)
- Bluetooth Mesh
- LoRa Mesh (for long-range IoT)
- **Mobile Ad Hoc Networks (MANETs)** – dynamic mesh for mobile nodes
 - Vehicular Ad hoc Network (VANETs)
 - Smart Phone Ad hoc Network (SPANC)
 - Internet based Mobile Ad hoc Network (iMANETs)
 - Hub-Spoke MANET
 - Military or Tactical MANETs
 - Flying Ad hoc Network (FANETs)

REAL-WORLD USE CASES OF WMN

- **Military & Tactical Networks**

- Troops carry nodes to form a mobile communication network in the field.

- **Disaster Recovery**

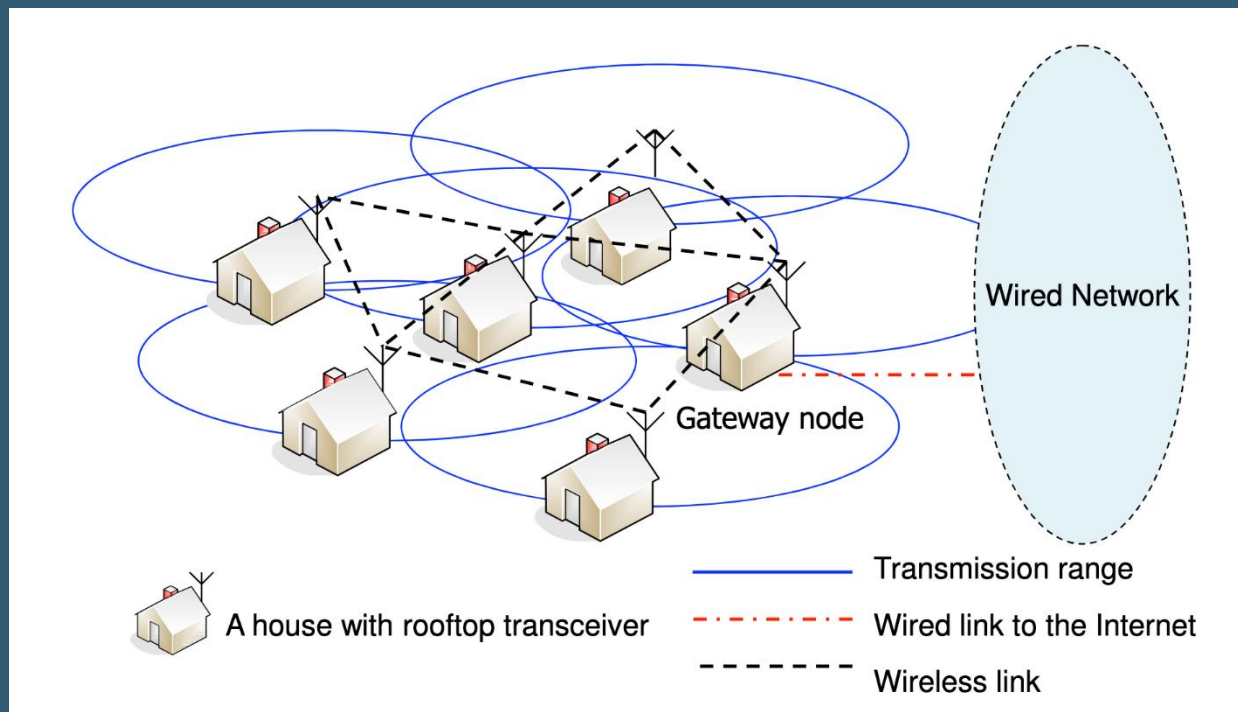
- Mesh networks set up rapidly after disasters (e.g., hurricanes, earthquakes) when infrastructure is down.

- **Industrial IoT**

- Sensor networks in factories, pipelines, or smart cities.

REAL-WORLD USE CASES OF WMN

- **Community Wi-Fi**
 - Mesh routers installed on buildings share internet across a neighbourhood.

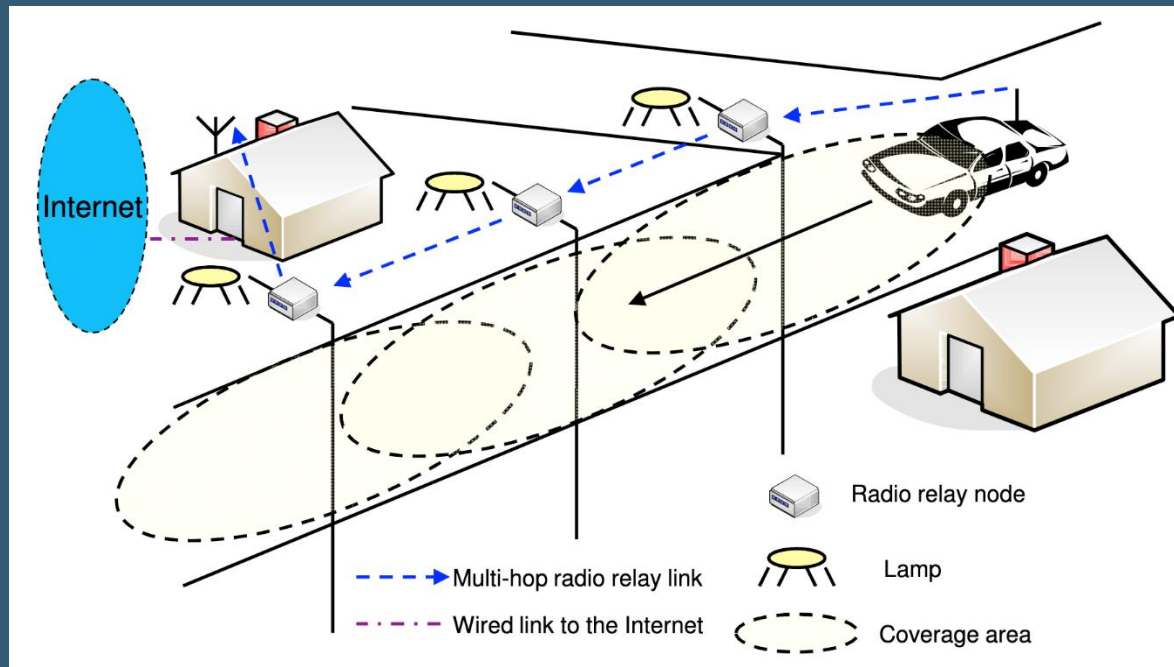


Wireless mesh networks operating in a residential zone

REAL-WORLD USE CASES OF WMN

- **Transportation**

- Buses or trains use mesh networks for vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication.



Wireless mesh network covering a highway

ROUTING PROTOCOL REQUIREMENTS FOR WMN

➤ Why is it different from routing in other types of network?

Because both end nodes and routers are mobile

- Self starting and self organizing
- Multi-hop operation with a routing mechanism designed for mobile nodes
- Dynamic topology maintenance
- Rapid convergence
- Minimal network traffic overhead
- Scalable to large networks

MOBILE AD HOC NETWORKS (MANET)

- The Internet Engineering Task Force (IETF) created the Mobile Ad-hoc Networks working group.
- The purpose of the MANET working group is to standardize IP routing protocol functionality suitable for wireless routing applications within dynamic topologies with increased dynamics due to node motion or other factors.
- The working group also serve as a meeting place and forum for those developing and experimenting with MANET approaches.
- Currently the group is pursuing a reactive, a proactive and hybrid protocol. No protocol has been standardized yet.

MANET ROUTING PROTOCOLS

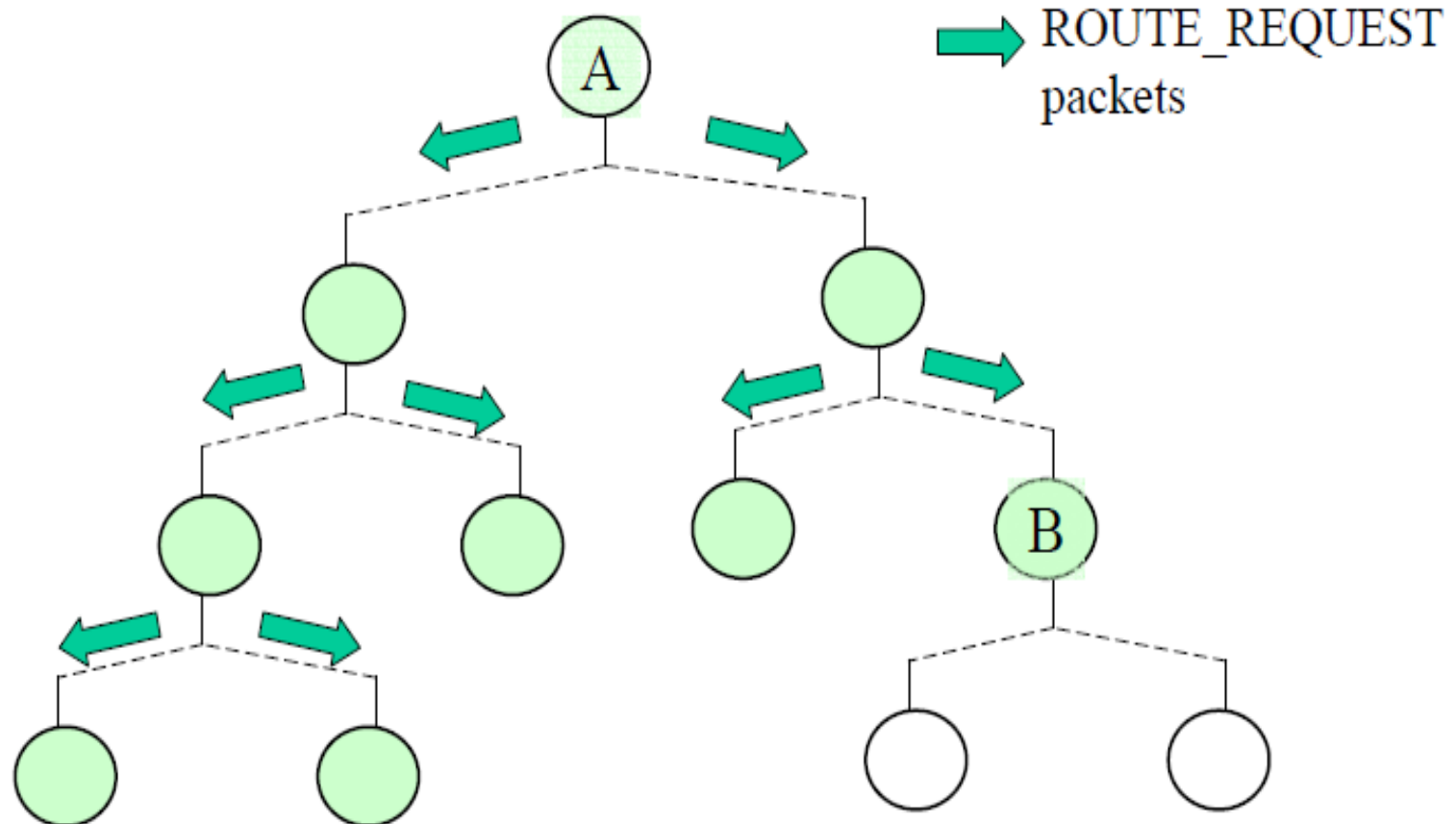
- Reactive

- Does not take initiative for finding routes
- Establishes routes “on demand” by flooding a query
- Less routing overhead in average because, does not use bandwidth except when needed (when finding a route)
- Much network overhead in the flooding process when querying for routes
- Higher latency in establishing the path
- Example: Dynamic Source Routing (DSR)

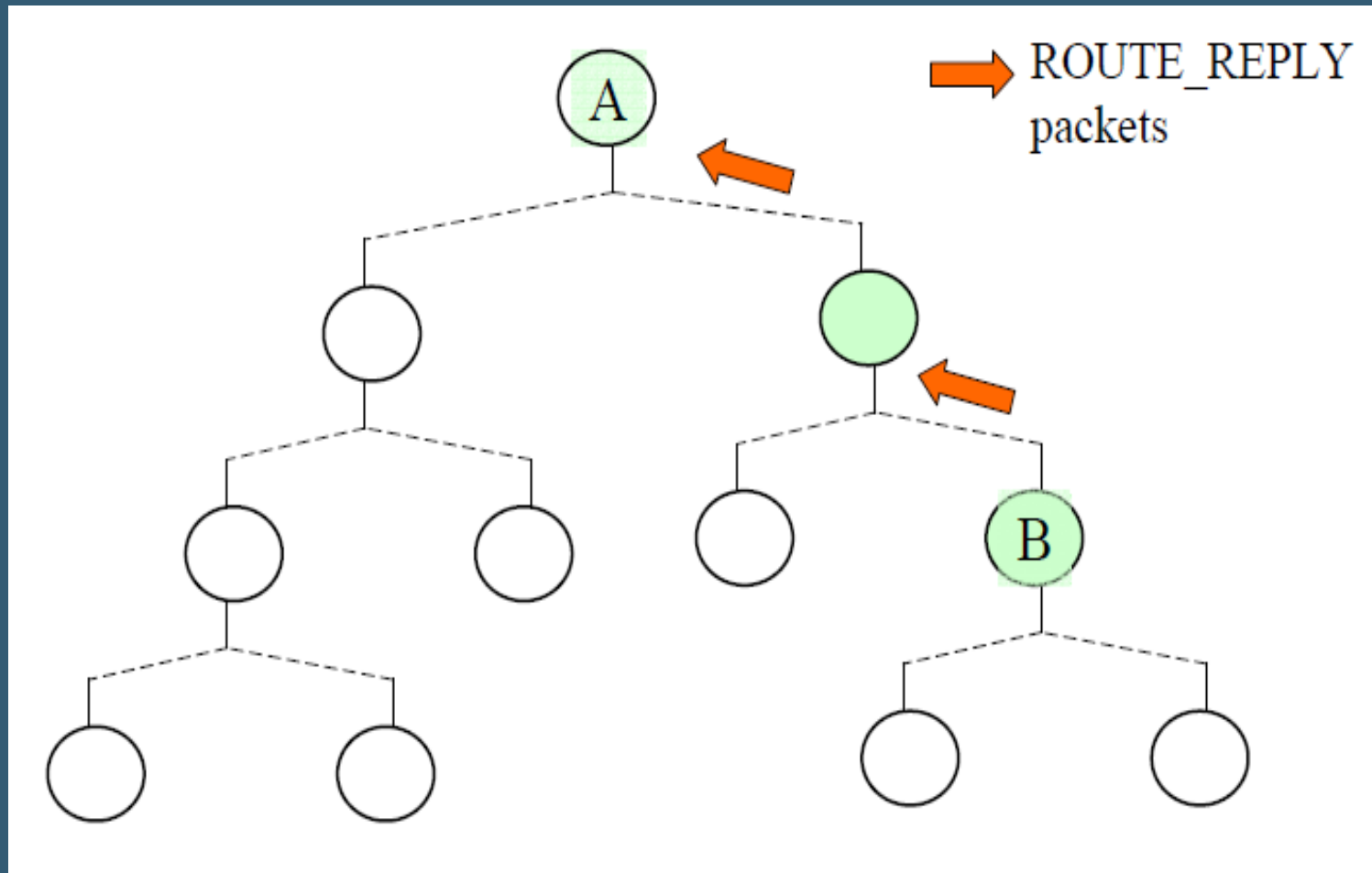
DRS PROTOCOL

- Suppose node A wishes to send a packet to node B, but does not currently have a valid route to the destination
 - Need for route discovery
- Node A broadcasts a ROUTE_REQUEST packet
 - Each node forwards the packet to its neighbors unless they are the destination or have a valid route to the destination
 - As the packet traverses the network, each intermediate node adds its address to the header, establishing the reverse route
- The destination, node B, sends a ROUTE_REPLY packet to node A
 - If the links are not bi-directional, node B must perform its own route discovery to respond to node A

DRS PROTOCOL



DRS PROTOCOL



DRS PROTOCOL

- Intermediate nodes may cache accumulated route record contained in the ROUTE_REQUEST packet headers in order to reduce routing overhead
 - Security concerns
- Confirmation of the receipt of a packet can be done by passive acknowledgement
 - Node overhears a downstream node forwarding the packet (Implicit ACK)
- DSR also contains provisions to avoid route reply storms
 - A network problem that occurs when multiple nodes simultaneously send Route Reply (RREP) messages in response to a Route Request (RREQ) in reactive routing protocols

PASSIVE ACKNOWLEDGMENT

- Passive acknowledgment means a node monitors the wireless medium to overhear its next-hop neighbour forwarding the packet it just sent.
- If it hears the packet being forwarded, it assumes it was successfully received.
- No explicit ACK is required - saving bandwidth and reducing overhead.
- For example there are three nodes A -> B -> C
 - **Node A** sends a packet to **Node B**.
 - **Node A** listens to the channel to **overhear** whether **Node B** forwards that packet to **Node C**.
 - If **Node A** overhears the packet from **Node B**, it considers this a **passive acknowledgment** that **Node B** received the packet successfully.
 - If A **does not overhear** the forwarding within a certain time, it may assume failure and try retransmitting or route error handling.

ROUTE REPLY STORMS

- A **route reply storm** occurs when:
 - Many nodes receive the same RREQ and several of them **have a route to the destination** in their **route cache**.
 - They all **respond at nearly the same time** with RREPs.
 - These simultaneous replies cause:
 - **Network congestion**
 - **Collisions at the MAC layer**
 - **Wasted bandwidth and energy**
- This is especially problematic in **dense networks** or when **route caching is aggressive** (as in DSR).

SOLUTIONS FOR ROUTE REPLY STORMS

- **Route Cache Suppression (DSR)**
 - Limit how often cached routes are used for RREPs.
 - Prevent multiple RREPs for the same RREQ from one node.
- **Random Delay Before RREP**
 - Add a **small random delay** before sending RREPs to reduce simultaneous replies.
- **RREP Rate Limiting**
 - Limit how many RREPs a node can send in a short time.
- **Duplicate RREP Suppression**
 - Ignore duplicate RREPs for the same RREQ from multiple nodes.
- **Use of Unicast for RREP**
 - Send RREPs via unicast rather than broadcast to reduce interference.

OLSR PROTOCOL

- The Optimized Link-State Routing protocol is a proactive based routing algorithm
- OLSR can be divided in to three main modules:
 - Neighbor/link sensing
 - Optimized flooding/forwarding (Multi Point Relaying)
 - Link-State messaging and route calculation

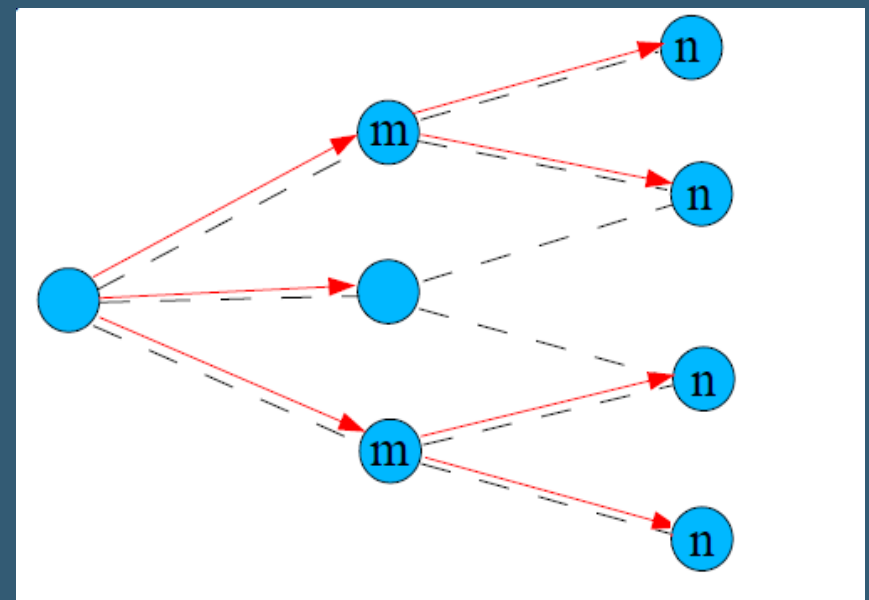
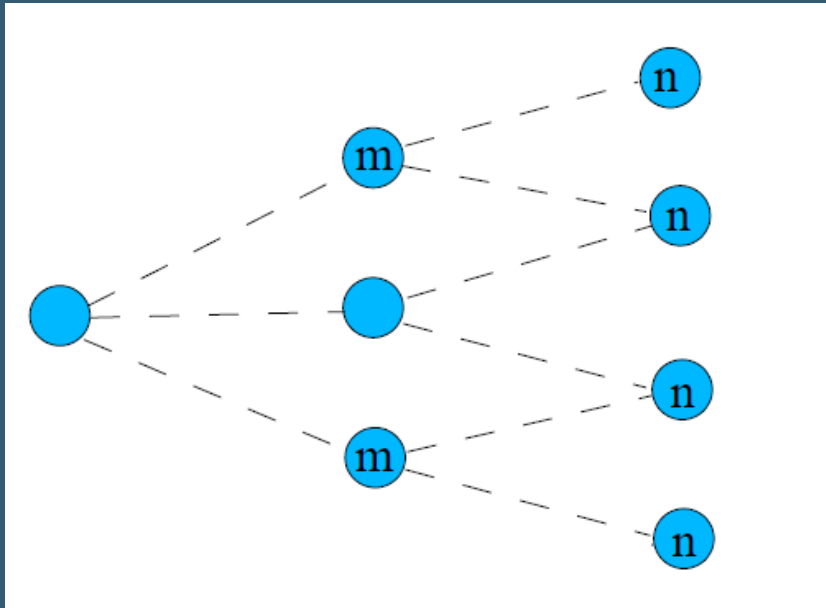
NEIGHBOR / LINK SENSING

- Routers maintain awareness of current network topology by exchanging “HELLO messages”
- All nodes transmit HELLO messages on a given interval.
- Each node tells the entire network about its immediate neighbors
 - So each node forms a picture of the entire network topology
 - Each node can then calculate the best route to any destination
- These contain all heard-of neighbors grouped by status.

MULTI-POINT RELAYING

- Flooding the network with HELLO messages incurs too much overhead
 - OLSR uses multi-point relay (MPR) nodes to decrease the number of unnecessary broadcasts (only selected nodes broadcast HELLO)
- Reduce the number of duplicate retransmissions while forwarding a broadcast packet.
- Restricts the set of nodes retransmitting a packet from all nodes(regular flooding) to a subset of all nodes.
- The size of this subset depends on the topology of the network.

MULTI-POINT RELAYING



LINK STATE FUNCTIONALITY

- In a classic link-state scheme all nodes flood the network with link-state information.
- OLSR has two link-state optimizations:
 - Only MPR selectors are declared in link-state messages. This minimizes the size of link-state messages.
 - Only nodes selected as MPRs will generate link-state messages. This minimizes the set of nodes emitting link-state messages.

MANET ROUTING PROTOCOLS

- Hybrid
 - This type of protocols combines the advantages of proactive and of reactive routing.
 - The routing is initially established with some proactively prospected routes and then serves the demand from additionally activated nodes through reactive flooding.
 - Advantage depends on number of nodes activated.
 - Example: Zone Routing Protocol (ZRP)

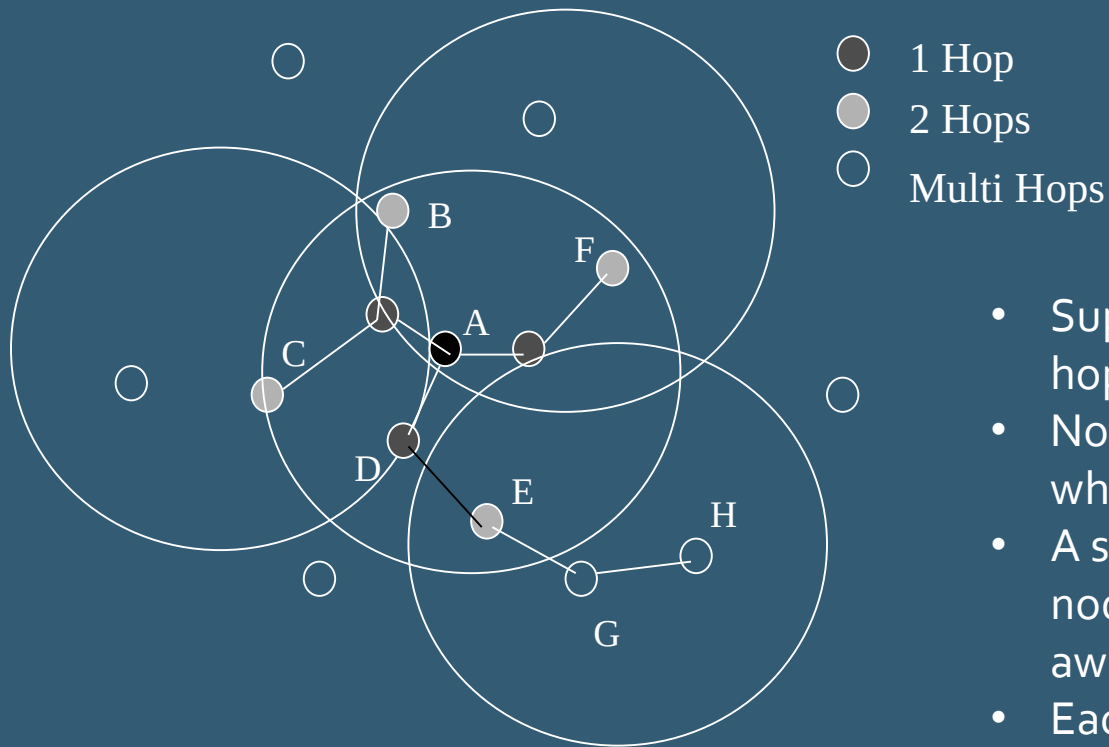
ZONE ROUTING PROTOCOL (ZRP)

- Proactive within the node's local neighborhood, reactive for inter-zone routing
 - Intra-zone routing: Proactively maintain routes to all nodes within the source node's own zone.
 - Inter-zone routing: Use an on-demand protocol (similar to DSR or AODV) to determine routes to outside zone.
- Proposed to reduce the control overhead of proactive routing protocols and decrease the latency caused by route discovery in reactive routing protocols
- Uses 'Bordercast' instead of neighbor broadcast
- Neighbor Discovery/Maintenance (NMD) and Border Resolution Protocol (BRP) used for query control, route accumulation etc.

BORDER RESOLUTION PROTOCOL (BRP)

- BRP is Used in **Zone Routing Protocol (ZRP)**, a **hybrid routing protocol** for MANETs.
- Each node maintains a **zone** (local neighborhood within k hops).
- For destinations **outside its zone**, it needs a mechanism to query efficiently, that's where **BRP** comes in.
 - Node wants to send to a destination **outside its zone**.
 - It generates a **Route Query Packet (RQP)**.
 - Instead of flooding the whole network, the RQP is **forwarded only to border nodes** (nodes exactly k hops away $\rightarrow$ "periphery of the zone").
 - Border nodes continue the query into their zones, and so on.
 - This reduces **broadcast overhead** compared to blind flooding.

ZONE ROUTING PROTOCOL (ZRP)



- Suppose $k = 2$ (zone radius = 2 hops).
- Node A wants to find Node H, which is outside its 2-hop zone.
- A sends query only to its border nodes B, F, E, C (which are at 2 hops away).
- Each border nodes check their zones; if H is found, reply is sent back.
- If not, they forward to their own border nodes.

MORE AD HOC ROUTING PROTOCOLS...

- ❑ Temporally Ordered Routing Algorithm (TORA)
 - Linked Cluster Architecture (LCA)
 - Reliable Ad hoc On-demand Distance Vector Routing Protocol
 - Ad hoc On-demand Routing Protocol (AORP)
 - Hybrid Routing Protocol for Large Scale Mobile Ad Hoc Networks with Mobile Backbones (HRPLS)
- ❑ Multicast routing protocols
 - Protocol Independent Multicast (PIM)
 - Multicast Zone Routing (MZR)
 - Multicast Optimized Link State Routing (MOLSR)
 - On-demand Multicast Routing Protocol (OMRP)